Modelling Moon Phases · Knowledge Organiser

Learning goal: I can test a 3D model of the Moon's orbit to record all eight phases and evaluate the model.

Today I Can

I can map each part of the torch, observer, and ball model to the Sun, Earth, and Moon.

I can record the 8 Moon phases in order from Position 1 (New Moon) to Position 8.

I can explain one strength and one limitation of our classroom Moon-phase model.

Must-Know Words

waxing crescent: the phase after New Moon when a thin sliver of light grows on the right

waxing gibbous: the phase before Full Moon when more than half of the face is lit and growing

waning gibbous: the phase after Full Moon when more than half is lit and shrinking

waning crescent: the final thin sliver of light before the cycle returns to New Moon

Core Science Steps

First, stand on the 8-point floor mat with Position 1 pointing straight toward the fixed lamp/torch.

Second, turn anticlockwise from Position 1 to Position 8, holding the ball at arm's length and observing only the side facing you.

Third, shade the dark fraction and leave the lit fraction white in each of the 8 phase circles in your book.

Keep the Science Clear

Record all 8 phases in order, then state one clear strength and one clear limitation of the physical model.

Classroom models cannot show true scale: the real Moon is about 30 Earth-diameters away from Earth, and the Sun is 400 times further still.